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Staffing problems with local network externalities[☆]

Yifan Xiong [*], Ziyan Li

School of Economics and Management, Anhui Normal University, Wuhu, CN, 241000, China

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ABSTRACT

This study examines how an organization assigns agents to interactive jobs embedded in a network. We employ a two-stage linear quadratic game with local complementarities, in which the underlying network represents the interrelated jobs that an organization seeks for staff. Under the assumption that employees have heterogeneous abilities, the organization assigns individuals to jobs in an attempt to maximize aggregate effort or aggregate payoff. Two problems are considered when maximizing the total effort. First, we assume that each agent's ability is single dimensional and show that the planner should match the ordinal ranking of an agent's ability level to the job indexed by the same ordinal level in the centrality ranking. Second, we assume that agents are experts in different jobs, implying that each agent's ability is multidimensional. We show that the optimal staffing problem is essentially an assignment problem that can be solved by the Hungarian algorithm in polynomial time. Finally, we consider an optimal staffing scenario to maximize the total payoff. Finding all permutations of n agents has high computation time complexity. To solve this problem, we design a heuristic algorithm.

1. Introduction

Numerous theoretical studies and empirical analyses have investigated peer effects or local network externalities in modern organizations. However, interactions are more dependent on an organization's structure than on the agents' social relations. For instance, the cooperation between a sales manager and sales assistant is due to the relevance of the two jobs, not their friendship. In contrast, even if they are privately good friends, a sales manager and a warehouseman can hardly directly influence each other at work. If we regard the organization's structure as a network, where jobs are the nodes and interactions are edges, optimal one-to-one matching of agents and nodes is an intriguing issue. In this study, we investigate this problem in several situations.

We now describe the main elements of the model. This study develops a two-stage network game with quadratic payoff, in which an organization first decides a staffing scenario that assigns agents to interactive jobs, whereby the agents then simultaneously select the effort level to maximize their payoff. The correlations between jobs form a complex network. Under this tractable framework, Ballester et al. (2006) showed that equilibrium actions are proportional to the Katz–Bonacich (KB) centralities Katz (1953), Bonacich (1987). When the organization wants to maximize agents' aggregate equilibrium effort, we find that agents should be ranked according to their ability, from high to low, and be one-to-one assigned to jobs from the most centralized node (measured by KB centralities when the network matrix is symmetric) to the least in the network.

In practice, people may be good at various jobs. We modeled this observation by extending each agent's ability to a vector, where the ith coefficient measures the agent's ability on the ith job. The staffing problem is equivalent to the classical assignment problem in this situation, which can be solved using the *Hungarian* algorithm in polynomial time (Kuhn, 1955).

If the interest of the organization depends on all agents' welfare, maximizing the aggregate equilibrium payoff is a more appropriate goal. Therefore, this study also examines the organization's optimal staffing scenario under this objective. However, the aggregate equilibrium payoff is intricate in the network configuration, whereas using the brute force algorithm requires computing $n!$ cases, which is not feasible when there are many agents. For this reason, we design a heuristic algorithm to approximately solve the problem instead of obtaining an exact solution. Simulation results show that the algorithm performed well.

Our staffing problems are related to the optimal sequential launch with network externalities that determine how to arrange

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[*] Corresponding author.
E-mail addresses: yifan@ahnu.edu.cn (Y. Xiong), lzy120@ahnu.edu.cn (Z. Li).

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agents to enter the network sequentially (See Berstein and Winter, 2012; Ajorlou et al., 2018; Swapna et al., 2012; Zhou and Chen, 2018). In previous studies, player positions were fixed in the network, whereas the present study considers how to match players and positions. To the best of our knowledge, this is a novel problem in network game research.

The remainder of this paper is organized as follows. In Section 2, we present the model setup, characterize the equilibrium, and study two staffing problems for maximizing the aggregate equilibrium effort. We then consider another objective maximizing total welfare in Section 3. Section 4 provides concluding remarks. Please refer to the appendix for all technical proofs.

2. The model

An organization of n agents exists in set $\mathcal{K}$ and n jobs that interact in set $\mathcal{I}$. We use the matrix $\mathbf{G} = [g_{ij}]$ to measure the interactions among different jobs. We assume that there are only positive local network externalities, that is, $g_{ij} > 0$ if job j influences job i; otherwise, $g_{ij} = 0$. Representative agent k's ability is denoted by θ_k ($\theta_k > 0\ \forall k \in \mathcal{K}$). Consider the following two-stage game:

Stage 1: The organization selects a staffing scenario $\psi : \mathcal{K} \to \mathcal{I}$, which is a one-to-one mapping of $\mathcal{I}$ onto $\mathcal{K}$ that matches all agents and jobs ($\psi(k) = i$ means agent k is assigned to jobs i). We denoted the set ψ as Ψ. Its cardinality $|\Psi|$ equals $n!$.

Stage 2: Under ψ, each agent $\psi^{-1}(i)$ simultaneously chose an effort $x_i \geqslant 0$ to maximize the payoff

$$y_i(x_1, \dots, x_n) = \theta_{\psi^{-1}(i)} x_i + \sum_{j \neq i} g_{ij} x_i x_j - \frac{1}{2} x_i^2. \tag{1}$$

Eq. (1) can be interpreted in the following manner: $\theta_{\psi^{-1}(i)} x_i$ is the output of position i by agent $\psi^{-1}(i)$'s own effort, $g_{ij} x_i x_j$ measures the output of position i because of agent $\psi^{-1}(j)$'s collaboration, and $\frac{1}{2} x_i^2$ is the cost of agent $\psi^{-1}(i)$ by exerting x_i. Let $\rho(\mathbf{G})$ denote the spectral radius $\mathbf{G}$. We introduce Assumption 1, which ensures the existence and uniqueness of an equilibrium.

Assumption 1. $\rho(\mathbf{G}) < 1$.

Proposition 1. *Suppose Assumption 1 holds and given a staffing scenario ψ, the simultaneous move of the n-player game with payoffs (1) and strategic space $\mathbb{R}_+$ has a unique Nash equilibrium, which is interior and given by*

$$\mathbf{x}^*(\mathbf{G}, \psi) = (\mathbf{I}_n - \mathbf{G})^{-1}\, \Theta[\psi], \tag{2}$$

where $\Theta[\psi] = \left(\theta_{\psi^{-1}(1)}, \dots, \theta_{\psi^{-1}(n)}\right)^\top$.

The proof of Proposition 1 refers to Ballester et al. (2010), so we omitted it here. Let $\mathbf{M} = [m_{ij}] = (\mathbf{I}_n - \mathbf{G})^{-1}$; then by the $\mathcal{M}$-matrix theory, elements m_{ij} of $\mathbf{M}$ are all nonnegative when Assumption 1 holds (see the survey of $\mathcal{M}$-matrices by Poole and Boullion, 1974). Hence, the unique equilibrium is also interior ($x_i^*(\mathbf{G}, \psi) \in \mathbb{R}_+$ for all i).

2.1. The optimal staffing

In this section, we assumed that the organization benefits τ from the agent's per unit of effort. By backward induction, the organization's first-stage optimization problem is

$$\mathcal{E} : \max_{\psi \in \Psi} \ \tau \cdot \sum_{i=1}^{n} x_i^*(\mathbf{G}, \psi). \tag{3}$$

Let $s_i(\mathbf{G}) = \sum_{j=1}^n m_{ji}$, problem $\mathcal{E}$ is equivalent to

$$\max_{\psi \in \Psi} \tau \cdot \sum_{i=1}^{n} \theta_{\psi^{-1}(i)} s_i(\mathbf{G}). \tag{4}$$

By ranking all $i \in \mathcal{I}$ and $k \in \mathcal{K}$ that satisfy $s_{i^1}(\mathbf{G}) \geqslant \cdots \geqslant s_{i^n}(\mathbf{G})$ and $\theta_{k^1} \geqslant \cdots \geqslant \theta_{k^n}$, we obtain two sequences, $i^1, \dots, i^n$ and $k^1, \dots, k^n$. When $\tau > 0$, the optimal staffing problem simply multiplies two groups of real numbers and maximizes the sum of the products. We obtain the following result.

Remark 1. Suppose Assumption 1 holds and $\tau > 0$, then the optimal staffing scenario ψ^E that solves problem $\mathcal{E}$ must satisfy $\psi^E(k^t) = i^t\ \forall t = 1, \dots, n$.[1]

Remark 1 shows that agents' abilities should match the importance of jobs, measured by the indices $s_i(\mathbf{G})$ in the network. Given a parameter δ and $\delta \cdot \rho(\mathbf{G}) < 1$, the variable $\mathbf{b}(\mathbf{G}, \delta) = (\mathbf{I}_n - \delta\mathbf{G})^{-1}\mathbf{1}_n$ is called the KB centrality. The ith element of $\mathbf{b}(\mathbf{G}, \delta)$ is equal to $\sum_{j=1} m_{ij}$, implying that $s_i(\mathbf{G}) = b_i(\mathbf{G}, 1)$ when $\mathbf{G}$ is symmetric. Remark 1 also provides an interesting observation: the most capable agent occupying the most important position may exert the lowest effort and obtain the lowest payoff under the optimal staffing scenario. We illustrate this point below:

Example 1. The organization has three agents with $\theta_1 = \theta_3 = a$ and $\theta_2 = a + \frac{1}{2}$ ($a > 0$), and three jobs with the following network configuration:

$$\mathbf{G} = \begin{bmatrix} 0 & a & 0 \\ 0 & 0 & 0 \\ 0 & a & 0 \end{bmatrix}.$$

From certain calculations, $s_1(\mathbf{G}) = s_3(\mathbf{G}) = 1$ and $s_2(\mathbf{G}) = 1 + 2a$. By Remark 1,

$$\psi^E(1) = 1,\ \psi^E(2) = 2,\ \psi^E(3) = 3,$$

and

$$x_1^*(\mathbf{G}, \psi^E) = x_3^*(\mathbf{G}, \psi^E) = a^2 + \frac{3}{2}a,\ \ x_2^*(\mathbf{G}, \psi^E) = a + \frac{1}{2}.$$

Because $x_2^*(\mathbf{G}, \psi^E) < x_1^*(\mathbf{G}, \psi^E)$ when $a > 0.25\left(\sqrt{3/2} - 1\right)$, Agent 2's equilibrium effort is the lowest. However, she/he is the most capable of all three agents. Furthermore, because $y_2^*(\mathbf{G}, \psi^E) = \frac{1}{2} x_2^{*2}(\mathbf{G}, \psi) < \frac{1}{2} x_1^{*2}(\mathbf{G}, \psi) = y_1^*(\mathbf{G}, \psi^E)$, when $a > 0.25\left(\sqrt{3/2} - 1\right)$, the equilibrium payoff of the most capable agent is also the lowest.

2.2. Multi-dimensional ability

In reality, people may be more suitable for different jobs. Let us now assume that each agent's ability is a vector of various jobs, that is, $\theta_k = (\theta_{k1}, \dots, \theta_{kn})$. Coefficient θ_{ki} measures agent k's ability to perform job i. Thus, the organization's optimization problem is:

$$\mathcal{E}^r : \ \max_{\psi \in \Psi} \sum_{i=1}^{n} \theta_{\psi^{-1}(i)i} \cdot s_i(\mathbf{G}).$$

Compared to the one-dimensional ability case, we should further consider players' comparative advantage in this context. For instance, suppose that $\theta_1 = (1, 9)$, $\theta_2 = (3, 10)$, $s_1(\mathbf{G}) = 1$, and

[1] Another consideration is how to optimally match agents and jobs when $\tau < 0$ (e.g. agents' actions are corrupt behaviors that are highly influenced by their colleagues' actions). In this case, the optimal staffing scenario minimizes the sum of the products of the two groups of real numbers. Evidently, $\psi^E(k^t) = i^{n-t}\ \forall t = 1,\ ldots, n$.

$s_2(\mathbf{G}) = 1.5$. Agent 2 is more capable than Agent 1 for job 2 because $\theta_{22} > \theta_{12}$. However,

$$\theta_{11}s_1(\mathbf{G}) + \theta_{22}s_2(\mathbf{G}) < \theta_{21}s_1(\mathbf{G}) + \theta_{12}s_2(\mathbf{G}),$$

which implies that Agent 2 should be assigned job 1. In fact, Agent 2 is more capable than Agent 1 for both jobs 1 and 2, while his/her advantage for job 1 is higher ($\theta_{21} - \theta_{11} > \theta_{22} - \theta_{12}$). Thus, Agents 1 and 2 have comparative advantages over jobs 2 and 1, respectively.

When there are more than two agents, comparative advantages are more complicated. The binary variables $x_{ij} \in \{0, 1\}$, $i, j = 1, \ldots, n$ correspond to the ijth event of assigning Agent j to job i ($x_{ij} = 1$ if Agent j is assigned to job i and $x_{ij} = 0$ otherwise). We obtain the following integer programming problem:

$$\begin{aligned} \mathcal{E}^r: \quad & \max \sum_{j=1}^{n}\sum_{i=1}^{n} c_{ij}\cdot x_{ij} \\ s.t \quad & \sum_{i=1}^{n} x_{ij} = 1 \\ & \sum_{j=1}^{n} x_{ij} = 1 \end{aligned}$$

where $c_{ij} = \theta_{ij}s_i(\mathbf{G})$. Given that network $\mathbf{G}$, $s_i(\mathbf{G})$ is constant for each i. Therefore, problem $\mathcal{E}^r$ is an assignment problem that can be solved by the Hungarian algorithm in polynomial time (see *integer and combinatorial optimization* by George and Laurence for details Kuhn, 1955).

Proposition 2. *Problem $\mathcal{E}^r$ can be solved in polynomial time.*

Note that each job's contribution to the total equilibrium effort is separable. This study has several policy implications.

Remark 2. Suppose some agents and jobs are fixed, denoting the set of all fixed jobs as $\mathcal{F}$, then the aggregate equilibrium efforts are

$$\sum_{i\in\mathcal{F}} \theta_{\psi^{-1}(i)i}\cdot s_i(\mathbf{G}) + \sum_{i\in\mathcal{I}|\mathcal{F}} \theta_{\psi^{-1}(i)i}\cdot s_i(\mathbf{G}).$$

Maximizing total effort is equivalent to maximizing $\sum_{i\in\mathcal{I}|\mathcal{F}} \theta_{\psi^{-1}(i)i}\cdot s_i(\mathbf{G})$. So Proposition 2 holds true.

3. To maximize the aggregate equilibrium payoff

This section investigates another surplus-sharing approach between organizations and agents. The social surplus created by job i is $y_i(x_1, \ldots, x_n)$ when all the agents' actions are $x_1, \ldots, x_n$. Suppose the organization takes $\upsilon \in (0, 1)$ shares of social surplus; then, each agent i's payoff is $(1-\upsilon)y_i(x_1, \ldots, x_n)$, and the organization's payoff is $\upsilon\cdot\sum_{i=1}^{n} y_i(x_1, \ldots, x_n)$. Agents' equilibrium actions are still characterized by Eq. (2) because υ is constant. By backward induction, the organization's optimization problem is:

$$\mathcal{P}: \max_{\psi\in\Psi} \sum_{i=1}^{n} y_i^*(\mathbf{G}, \psi). \tag{5}$$

The first-order condition of agent $\psi^{-1}(i)$ is:

$$\theta_{\psi^{-1}(i)} + \sum_{j\neq i} g_{ij}x_j^*(\mathbf{G}, \psi) - x_i^*(\mathbf{G}, \psi) = 0.$$

From Proposition 1, we have

$$\begin{aligned} y_i^*(\mathbf{G}, \psi) &= \left(\theta_{\psi^{-1}(i)} + \sum_{j\neq i} g_{ij}x_j^*(\mathbf{G}, \psi) - \frac{1}{2}x_i^*(\mathbf{G}, \psi)\right)x_i^*(\mathbf{G}, \psi) \\ &\quad + \frac{1}{2}x_i^*(\mathbf{G}, \psi)^2 \\ &= \frac{1}{2}x_i^*(\mathbf{G}, \psi)^2 \end{aligned}$$

and $\sum_{i=1}^{n} y_i^*(\mathbf{G}, \psi) = \frac{1}{2}\mathbf{x}^*(\mathbf{G}, \psi)^\top\mathbf{x}^*(\mathbf{G}, \psi) = \Theta[\psi]^\top\mathbf{M}^\top\mathbf{M}\Theta[\psi]$. Problem $\mathcal{P}$ can be rewritten as:

$$\mathcal{P}: \max_{\psi\in\Psi} \Theta[\psi]^\top\mathbf{M}^\top\mathbf{M}\Theta[\psi].$$

Problem $\mathcal{P}$ is intractable owing to the intricate structure of the aggregate equilibrium payoff. The time complexity of the brute-force algorithm that calculates all permutations of n agents is $\mathcal{O}(n\cdot n!)$, which is unsolvable within a reasonable time with a large n.[2] Thus, it is suitable to use algorithmic approximations to solve problem $\mathcal{P}$. First, we introduce some notations to facilitate the discussion.

When $\rho(\mathbf{G}) < 1$:

1. Let ψ^P denote the solution of problem $\mathcal{P}$, which is assumed unique for simplicity.
2. For any $\mathcal{M} \subseteq \mathcal{K}$, $\psi_{\mathcal{M}}: \mathcal{M} \to \mathcal{I}$ is a one-to-one mapping (may not be a surjection) of $\mathcal{M}$ onto $\mathcal{I}$ that assigns the agents in set $\mathcal{M}$ to the set of jobs.
3. Staffing scenario $\psi_{\mathcal{M}}^P$ is the one-to-one mapping of $\mathcal{M}$ onto $\mathcal{I}$ that satisfies $\psi_{\mathcal{M}}^P(k) = \psi^P(k)\ \forall k \in \mathcal{M}$.
4. The set $\mathcal{I}_{\psi_{\mathcal{M}}} = \{i \in \mathcal{I} | i = \psi_{\mathcal{M}}(k), \forall k \in \mathcal{M}\}$ includes all jobs that match the agents in the set $\mathcal{M}$.
5. Let $\mathbf{x}^{-\mathcal{I}_{\psi_{\mathcal{M}}}}$ denote the $n - |\mathcal{I}_{\psi_{\mathcal{M}}}|$ dimensional vector of x_i, $i \in \mathcal{I}|\mathcal{I}_{\psi_{\mathcal{M}}}$, and
$$F\left(\mathbf{x}^{-\mathcal{I}_{\psi_{\mathcal{M}}}}, \psi_{\mathcal{M}}\right) = \frac{1}{2}\times\left\{\begin{array}{c} \sum_{j\in\mathcal{I}_{\psi_{\mathcal{M}}}}\sum_{i\in\mathcal{I}_{\psi_{\mathcal{M}}}} \hat{m}_{ij}\theta_{\psi_{\mathcal{M}}^{-1}(i)}\theta_{\psi_{\mathcal{M}}^{-1}(j)} \\ +2\sum_{j\in\mathcal{I}_{\psi_{\mathcal{M}}}}\sum_{i\notin\mathcal{I}_{\psi_{\mathcal{M}}}} \hat{m}_{ij}x_i\theta_{\psi_{\mathcal{M}}^{-1}(j)} \\ +\sum_{j\notin\mathcal{I}_{\psi_{\mathcal{M}}}}\sum_{i\notin\mathcal{I}_{\psi_{\mathcal{M}}}} \hat{m}_{ij}x_ix_j \end{array}\right\},$$
which is a quadratic function of $\mathbf{x}^{-\mathcal{I}_{\psi_{\mathcal{M}}}}$. The matrix form of $F\left(\mathbf{x}^{-\mathcal{I}_{\psi_{\mathcal{M}}}}, \psi_{\mathcal{M}}\right)$ is
$$F\left(\mathbf{x}^{-\mathcal{I}_{\psi_{\mathcal{M}}}}, \psi_{\mathcal{M}}\right) = \frac{1}{2}\cdot\mathbf{z}^\top\hat{\mathbf{M}}\mathbf{z},$$
where $z_i = \theta_{\psi_{\mathcal{M}}^{-1}(i)}$ for all $i \in \mathcal{I}_{\psi_{\mathcal{M}}}$ and $z_i = x_i$ for all $i \notin \mathcal{I}_{\psi_{\mathcal{M}}}$.

Because $\hat{m}_{ij}$ and θ_k are positive for all i, j, k, we can easily show that

$$\begin{aligned} F\left(\min_{k\in\mathcal{K}|\mathcal{M}} \theta_k\cdot\mathbf{1}_{(n-|\mathcal{M}|)}, \psi_{\mathcal{M}}^P\right) &< \frac{1}{2}\Theta[\psi^P]^\top\hat{\mathbf{M}}\Theta[\psi^P] \\ &< F\left(\max_{k\in\mathcal{K}|\mathcal{M}} \theta_k\cdot\mathbf{1}_{(n-|\mathcal{M}|)}, \psi_{\mathcal{M}}^P\right), \end{aligned}$$

where $\mathbf{1}_{n-|\mathcal{M}|}$ is the $n-|\mathcal{M}|$-dimensional vector whose elements are all equal to one. Furthermore, there exists a constant $\alpha_{\mathcal{M}} \in \left(\min_{k\in\mathcal{K}|\mathcal{M}}\theta_k, \max_{k\in\mathcal{K}|\mathcal{M}}\theta_k\right)$ such that

$$F\left(\alpha_{\mathcal{M}}\cdot\mathbf{1}_{(n-|\mathcal{M}|)}, \psi_{\mathcal{M}}^P\right) = \frac{1}{2}\Theta[\psi^P]^\top\hat{\mathbf{M}}\Theta[\psi^P].$$

The following proposition gives a lower bound of $\alpha_{\mathcal{M}}$.

Proposition 3. *Suppose Assumption 1 holds, $\alpha_{\mathcal{M}} > \frac{\sum_{k\in\mathcal{K}|\mathcal{M}}\theta_k}{n-|\mathcal{M}|}$.*

[2] Finding all permutations of n numbers can be decomposed into n subproblems of finding all permutations of $n-1$ numbers. Thus, to illustrate, all permutations of n numbers need to proceed with n steps by recursion. In the first step, there are n permutations. In the second step, there are $n-1$ permutations, and in each permutation, we call the result of the first step by operating n times. Therefore, there are $n\cdot(n-1)$ operations in the second step. By recursion, there are $n\cdot(n-1)\cdot\cdots\cdot(n-k+1)$ operations in Step k. In conclusion, the total number of operations of all steps is $n + n\cdot(n-1) + \cdots + n! < n\cdot n!$, implying that the time complexity of this recursion is $\mathcal{O}(n\cdot n!)$.

Proof. See Appendix.

This proposition inspired us to construct a program that iterates the assignment of agents to jobs. Let $\mathcal{K}^t = \{k^1, \ldots, k^t\}$, which is the set of the top k capable agents in $\mathcal{K}$. The staffing scenario $\psi_{\mathcal{K}^t}$ assigns agents in sets $\mathcal{K}^t$, and $\psi_{\{k^t\}}$ assigns the tth capable agent, so we can write $\psi_{\mathcal{K}^t}$ in a recursive form as $\psi_{\mathcal{K}^{t+1}} = \psi_{\mathcal{K}^t} \bigcup \psi_{\{k^{t+1}\}}$. Proposition 3 suggests Algorithm 1 to solve problem $\mathcal{P}$.

Algorithm 1 (*Compute the Optimal Staffing*)**.**

- Step 1.
  - Initialize $t = 1$ and pick $\hat{\alpha}_{\mathcal{K}^1} \in \left(\frac{1}{n-1} \cdot \sum_{k\in\mathcal{K}|\mathcal{K}^1} \theta_k, \max_{k\in\mathcal{K}|\mathcal{K}^1} \theta_k\right)$.
  - Let $\psi^{op}_{\mathcal{K}^1} = \arg\max_{\psi_{\mathcal{K}^1}} F\left(\hat{\alpha}_{\mathcal{K}^1} \cdot \mathbf{1}_{n-1}, \psi_{\mathcal{K}^1}\right)$.
- Step 2.
  - Pick $\hat{\alpha}_{\mathcal{K}^t} \in \left(\frac{1}{n-t} \cdot \sum_{k\in\mathcal{K}|\mathcal{K}^t} \theta_k, \max_{k\in\mathcal{K}|\mathcal{K}^t} \theta_k\right)$.
  - Let $\psi^{op}_{\mathcal{K}^t} = \arg\max_{\psi_{\mathcal{K}^t}} F\left(\hat{\alpha}_{\mathcal{K}^t} \cdot \mathbf{1}_{n-t}, \psi^{op}_{\mathcal{K}^{t-1}} \bigcup \psi_{\{k^t\}}\right)$.
  - **IF** $t < n$, return to Step 2 **ELSE IF Output** $\psi^{op} = \psi^{op}_{\mathcal{K}^n}$.

Algorithm 1 terminates when $t = n$ and obtains a staffing scenario ψ^{op} in which one-to-one matches the group of agents and jobs. The complexity of the algorithm is $O(n^2)$ because there are n steps, and each step involves a comparison of at most n cases. Example 2 illustrates the operation of the algorithm.

Example 2. Consider the case of four agents and four jobs. Assuming $\theta_1 = 1, \theta_2 = 2, \theta_3 = 3, \theta_4 = 4$,

$$\mathbf{G} = \begin{bmatrix} 0 & 0 & 0.3 & 0.3 \\ 0.3 & 0 & 0 & 0 \\ 0.3 & 0 & 0 & 0.3 \\ 0 & 0.3 & 0 & 0 \end{bmatrix}$$

and $\hat{\alpha}_{\mathcal{K}^t} = \sqrt{\sum_{k\in\mathcal{K}|\mathcal{K}^t} \theta_k^2/(n-t)}$, then $\hat{\alpha}_{\mathcal{K}^1} = \sqrt{\frac{1^2+2^2+3^2}{3}} = \sqrt{\frac{14}{3}}$, $\hat{\alpha}_{\mathcal{K}^2} = \sqrt{\frac{1^2+2^2}{2}} = \sqrt{\frac{5}{2}}$, and $\hat{\alpha}_{\mathcal{K}^3} = \sqrt{\frac{1^2}{1}} = 1$.[3] The first step of the algorithm is to compare

$$\left(4, \sqrt{\tfrac{14}{3}}, \sqrt{\tfrac{14}{3}}, \sqrt{\tfrac{14}{3}}\right) \hat{\mathbf{M}} \left(4, \sqrt{\tfrac{14}{3}}, \sqrt{\tfrac{14}{3}}, \sqrt{\tfrac{14}{3}}\right)^\top = 139.243,$$

$$\left(\sqrt{\tfrac{14}{3}}, 4, \sqrt{\tfrac{14}{3}}, \sqrt{\tfrac{14}{3}}\right) \hat{\mathbf{M}} \left(\sqrt{\tfrac{14}{3}}, 4, \sqrt{\tfrac{14}{3}}, \sqrt{\tfrac{14}{3}}\right)^\top = 122.864,$$

$$\left(\sqrt{\tfrac{14}{3}}, \sqrt{\tfrac{14}{3}}, 4, \sqrt{\tfrac{14}{3}}\right) \hat{\mathbf{M}} \left(\sqrt{\tfrac{14}{3}}, \sqrt{\tfrac{14}{3}}, 4, \sqrt{\tfrac{14}{3}}\right)^\top = 136.328,$$

$$\left(\sqrt{\tfrac{14}{3}}, \sqrt{\tfrac{14}{3}}, \sqrt{\tfrac{14}{3}}, 4\right) \hat{\mathbf{M}} \left(\sqrt{\tfrac{14}{3}}, \sqrt{\tfrac{14}{3}}, \sqrt{\tfrac{14}{3}}, 4\right)^\top = 145.289,$$

thus $\psi^{op}(4) = 4$. The second step of the algorithm is to compare

$$\left(3, \sqrt{\tfrac{5}{2}}, \sqrt{\tfrac{5}{2}}, 4\right) \hat{\mathbf{M}} \left(3, \sqrt{\tfrac{5}{2}}, \sqrt{\tfrac{5}{2}}, 4\right)^\top = 143.754,$$

$$\left(\sqrt{\tfrac{5}{2}}, 3, \sqrt{\tfrac{5}{2}}, 4\right) \hat{\mathbf{M}} \left(\sqrt{\tfrac{5}{2}}, 3, \sqrt{\tfrac{5}{2}}, 4\right)^\top = 131.514,$$

$$\left(\sqrt{\tfrac{5}{2}}, \sqrt{\tfrac{5}{2}}, 3, 4\right) \hat{\mathbf{M}} \left(\sqrt{\tfrac{5}{2}}, \sqrt{\tfrac{5}{2}}, 3, 4\right)^\top = 141.548,$$

thus $\psi^{op}(3) = 1$. The third step of the algorithm is to compare

$$(3, 2, 1, 4)\, \hat{\mathbf{M}}\, (3, 2, 1, 4)^\top = 137.014,$$

$$(3, 1, 2, 4)\, \hat{\mathbf{M}}\, (3, 1, 2, 4)^\top = 144.69,$$

thus $\psi^{op}(2) = 3$ and $\psi^{op}(1) = 2$. Above all, ψ^{op} assigns agents 1, 2, 3, 4 to positions 3, 1, 2, 4, respectively. Table 1 illustrates all possible staffing scenarios in Example 2 and the corresponding aggregated equilibrium payoff, which shows that ψ^{op} is the same as ψ^P in this example.

Generally, it is difficult to obtain the specific accuracy of a heuristic algorithm. Therefore, we evaluated the accuracy of Algorithm 1 using simulations. Given the network matrix $\mathbf{G}$ and all agents' abilities Θ, we let

$$\epsilon(\hat{\mathbf{G}}) = \frac{\sum_{i=1}^n y_i^*(\mathbf{G}, \psi^{op}) - \min_{\psi\in\Psi} \sum_{i=1}^n y_i^*(\mathbf{G}, \psi)}{\sum_{i=1}^n y_i^*(\mathbf{G}, \psi^P) - \min_{\psi\in\Psi} \sum_{i=1}^n y_i^*(\mathbf{G}, \psi)}.$$

The closer $\epsilon(\hat{\mathbf{G}})$ is to 1, the more accurate the algorithm (in particular, $\psi^{op} = \psi^P$ when $\epsilon(\mathbf{G}) = 1$). The real value of $\alpha_{\mathcal{K}^t}$ is unknown before obtaining ψ^P for each step t; therefore, we need to iteratively estimate $\alpha_{\mathcal{K}^t}$ in each step to perform the algorithm.

We define the *accuracy* of Algorithm 1 as the proportion of networks that satisfy $\epsilon(\mathbf{G}) = 1$ and

$$\eta(\mathbf{G}) = 1 - \epsilon(\mathbf{G}),$$

$$\hat{a}_1 = \sqrt{\frac{\sum_{k\in\mathcal{K}|\mathcal{K}^t} \theta_k^2}{n-t}},$$

$$\hat{a}_2 = 0.5 \cdot \max_{k\in\mathcal{K}|\mathcal{K}^t} \theta_k + 0.5 \cdot \frac{\sum_{k\in\mathcal{K}|\mathcal{K}^t} \theta_k}{n-t},$$

$$\hat{a}_3 = 0.75 \cdot \max_{k\in\mathcal{K}|\mathcal{K}^t} \theta_k + 0.25 \cdot \frac{\sum_{k\in\mathcal{K}|\mathcal{K}^t} \theta_k}{n-t},$$

$$\hat{a}_4 = 0.25 \cdot \max_{k\in\mathcal{K}|\mathcal{K}^t} \theta_k + 0.75 \cdot \frac{\sum_{k\in\mathcal{K}|\mathcal{K}^t} \theta_k}{n-t}.$$

The index $\eta(\mathbf{G})$ measures the deviation of Algorithm 1's result from the solution of problem $\mathcal{P}$, and $\hat{a}_i$ ($i = 1, 2, 3, 4$) are the four estimators of $\alpha_{\mathcal{K}^t}$. Table 2 presents the performance of Algorithm 1 by calculating 100 randomly generated networks for $n = 6, 7, 8, 9, 10$.

Table 2 shows that Algorithm 1 is effective (its accuracy is more than 90%). Even if Algorithm 1 fails to get the optimal solution, its deviation from the optimal solution is negligible. Furthermore, Algorithm 1 has no significant difference under four estimators, which implies it is also robust in solving problem $\mathcal{P}$.

4. Discussion

This study examined how an organization assigns agents to jobs embedded in a network. First, we characterized the staffing problems that maximized aggregate equilibrium efforts in two cases. We then explored a heuristic algorithm to search for an optimal staffing scenario that maximizes total welfare. A major contribution of this study was the development of a matching framework for network games.

Our study can naturally be extended to the case where the number of agents is greater than the number of jobs. Suppose n jobs and m agents exist, where $m > n$. When each agent's ability is one-dimensional, the organization should assign the top

[3] The square mean is always strictly greater than the arithmetic mean, so $\sqrt{\frac{\sum_{k\in\mathcal{K}|\mathcal{K}^t} \theta_k^2}{n-t}} \in \left(\frac{\sum_{k\in\mathcal{K}|\mathcal{K}^t} \theta_k}{n-t}, \max_{k\in\mathcal{K}|\mathcal{K}^t} \theta_k\right)$.

Table 1
All possible staffing scenarios in Example 3.

$\psi(k)$	$\Theta[\psi]^\top \hat{M}\Theta[\psi]$	$\psi(k)$	$\Theta[\psi]^\top \hat{M}\Theta[\psi]$	$\psi(k)$	$\Theta[\psi]^\top \hat{M}\Theta[\psi]$	$\psi(k)$	$\Theta[\psi]^\top \hat{M}\Theta[\psi]$
1,2,3,4	133.671	2,1,3,4	143.325	3,1,2,4	$\boxed{144.69}$	4,1,2,3	141.004
1,2,4,3	128.218	2,1,4,3	138.458	3,1,4,2	136.455	4,1,3,2	137.636
1,3,2,4	126.977	2,3,1,4	128.956	3,2,1,4	137.014	4,2,1,3	132.957
1,3,4,2	116.398	2,3,4,1	115.336	3,2,4,1	125.153	4,2,3,1	126.548
1,4,2,3	115.07	2,4,1,3	117.263	3,4,1,2	114.146	4,3,1,2	121.781
1,4,3,2	109.944	2,4,3,1	108.511	3,4,2,1	110.52	4,3,2,1	118.741

Table 2
The results of Algorithm 1 for 500 randomly generated networks.

	$\hat{a}_1$		$\hat{a}_2$		$\hat{a}_3$		$\hat{a}_4$	
	Accuracy	max $\eta(G)$	Accuracy	max $\eta(G)$	Accuracy	max $\eta(G)$	Accuracy	max $\eta(G)$
6	0.96	0.0117	0.95	0.0117	0.91	0.0117	0.92	0.0117
7	0.89	0.0587	0.89	0.0587	0.89	0.0587	0.89	0.0587
8	0.99	0.0001	0.99	0.0001	0.99	0.0001	0.99	0.0001
9	0.90	0.0024	0.90	0.0024	0.89	0.0024	0.89	0.0024
10	0.99	0.0001	0.99	0.0001	0.99	0.0001	0.99	0.0001

n capable agents to jobs according to our proposed methods, regardless of maximizing aggregate actions or payoffs. For the multidimensional ability case, we can imagine $m-n$ jobs with zero centralities ($s_i(\mathbf{G})=0$ for all $i \in \{n+1,\ldots,m\}$), and then solve an assignment problem with m agents and m jobs using the Hungarian algorithm. When the number of jobs is greater than the number of agents ($n > m$), the organization must determine which n jobs should be retained. The optimal staffing scenario for this situation remains an open question.

Throughout the study, we assumed that both the organization and agents know each agent's ability. However, obtaining information regarding the agents' abilities is a crucial problem. Extending our analysis by introducing incomplete information is an interesting direction for future research.

Appendix

Proof of Proposition 3. Since $\mathbf{M}$ is invertible when $\rho(\mathbf{G}) < 1$, for any non-zero vector $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{Mx} \neq 0$ and $\mathbf{x}^\top \mathbf{M}^\top \mathbf{Mx} = \|\mathbf{Mx}\|^2 > 0$. Hence, $\mathbf{M}^\top\mathbf{M} = [\hat{m}_{ij}]$ is positive definite, implying that all principal minors of $[\hat{m}_{ij}]$ are positive definite. Thus, the matrix $[\hat{m}_{ij}]^{-\mathcal{I}_{\psi_\mathcal{M}}}$, which eliminates the ith row and jth column from $[\hat{m}_{ij}]$ for all $i,j \in \mathcal{I}_{\psi_\mathcal{M}}$, is positive definite. The matrix $diag[\hat{m}_{ii}]^{-\mathcal{I}_{\psi_\mathcal{M}}}$, which eliminates the ith row and ith column from $diag[\hat{m}_{ii}]$ for all $i,j \in \mathcal{I}_{\psi_\mathcal{M}}$, is also positive definite. It follows that

$$\nabla_{\mathbf{x}^{-\mathcal{I}_{\psi_\mathcal{M}}}} F\left(\mathbf{x}^{-\mathcal{I}_{\psi_\mathcal{M}}}, \psi_\mathcal{M}\right) = [\hat{m}_{ij}]^{-\mathcal{I}_{\psi_\mathcal{M}}} + diag[\hat{m}_{ii}]^{-\mathcal{I}_{\psi_\mathcal{M}}}$$

is positive definite, and $F\left(\mathbf{x}^{-\mathcal{I}_{\psi_\mathcal{M}}}, \psi_\mathcal{M}\right)$ is a strictly convex function of $\mathbf{x}^{-\mathcal{I}_{\psi_\mathcal{M}}}$. Therefore, for all possible staffing scenarios, $\psi_{\mathcal{K}|\mathcal{M}} \in \Psi_{\mathcal{K}|\mathcal{M}}$ (the cardinality of $\Psi_{\mathcal{K}|\mathcal{M}}$ equals $(n-|\mathcal{M}|)!$),

$$\frac{1}{(n-|\mathcal{M}|)!} \cdot \sum_{\psi_{\mathcal{K}|\mathcal{M}} \in \Psi_{\mathcal{K}|\mathcal{M}}} F\left(\Theta[\psi_{\mathcal{K}|\mathcal{M}}], \psi_\mathcal{M}\right) > F\left(\frac{\sum_{\psi_{\mathcal{K}|\mathcal{M}} \in \Psi_{\mathcal{K}|\mathcal{M}}} \Theta[\psi_{\mathcal{K}|\mathcal{M}}]}{(n-|\mathcal{M}|)!}, \psi_\mathcal{M}\right), \tag{A.1}$$

where the coefficients of $n-|\mathcal{M}|$-dimensional vector $\Theta[\psi_{\mathcal{K}|\mathcal{M}}]$ are $\theta_k\ \forall k \in \mathcal{K}|\mathcal{M}$. Since

$$\sum_{\psi_{\mathcal{K}|\mathcal{M}} \in \Psi_{\mathcal{K}|\mathcal{M}}} \Theta[\psi_{\mathcal{K}|\mathcal{M}}] = (n-|\mathcal{M}|-1)! \cdot \sum_{k\in\mathcal{K}|\mathcal{M}} \theta_k \cdot \mathbf{1}^\top_{(n-|\mathcal{M}|)}$$

and

$$\frac{\sum_{\psi_{\mathcal{K}|\mathcal{M}} \in \Psi_{\mathcal{K}|\mathcal{M}}} \Theta[\psi_{\mathcal{K}|\mathcal{M}}]}{(n-|\mathcal{M}|)!} = \frac{\sum_{k\in\mathcal{K}|\mathcal{M}} \theta_k}{n-|\mathcal{M}|} \cdot \mathbf{1}^\top_{(n-|\mathcal{M}|)},$$

Eq. (A.1) can be rewritten as

$$\begin{aligned} F\left(\frac{\sum_{k\in\mathcal{K}|\mathcal{M}} \theta_k}{n-|\mathcal{M}|} \cdot \mathbf{1}^\top_{(n-|\mathcal{M}|)}, \psi_\mathcal{M}\right) &< \sum_{\psi_{\mathcal{K}|\mathcal{M}} \in \Psi_{\mathcal{K}|\mathcal{M}}} \frac{F(\Theta[\psi_{\mathcal{K}|\mathcal{M}}]; \psi_\mathcal{M})}{(n-|\mathcal{M}|)!} \\ &< \max_{\psi_{\mathcal{K}|\mathcal{M}} \in \Psi_{\mathcal{K}|\mathcal{M}}} \left\{F(\Theta[\psi_{\mathcal{K}|\mathcal{M}}]; \psi_\mathcal{M})\right\} \\ &= F\left(\alpha_\mathcal{M} \cdot \mathbf{1}^\top_{(n-|\mathcal{M}|)}, \psi_\mathcal{M}\right). \end{aligned}$$

As $\frac{\partial F\left(z\cdot\mathbf{1}^\top_{(n-|\mathcal{M}|)}, \psi_\mathcal{M}\right)}{\partial z} > 0$ when $z > 0$, we have $\alpha_\mathcal{M} > \frac{\sum_{k\in\mathcal{K}|\mathcal{M}} \theta_k}{n-|\mathcal{M}|}$. $\square$

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